```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.datasets import fetch_california_housing
```

```
housing = fetch_california_housing()
data = fetch_california_housing(as_frame=True).frame
```

```
data.head()
```

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24	3.521
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25	3.413
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25	3.422

Next steps: Generate code with data New interactive sheet

```
data.shape
```

```
(20640, 9)
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20640 entries, 0 to 20639
Data columns (total 9 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   MedInc       20640 non-null  float64
 1   HouseAge     20640 non-null  float64
 2   AveRooms     20640 non-null  float64
 3   AveBedrms    20640 non-null  float64
 4   Population   20640 non-null  float64
 5   AveOccup     20640 non-null  float64
 6   Latitude     20640 non-null  float64
 7   Longitude    20640 non-null  float64
 8   MedHouseVal  20640 non-null  float64
dtypes: float64(9)
memory usage: 1.4 MB
```

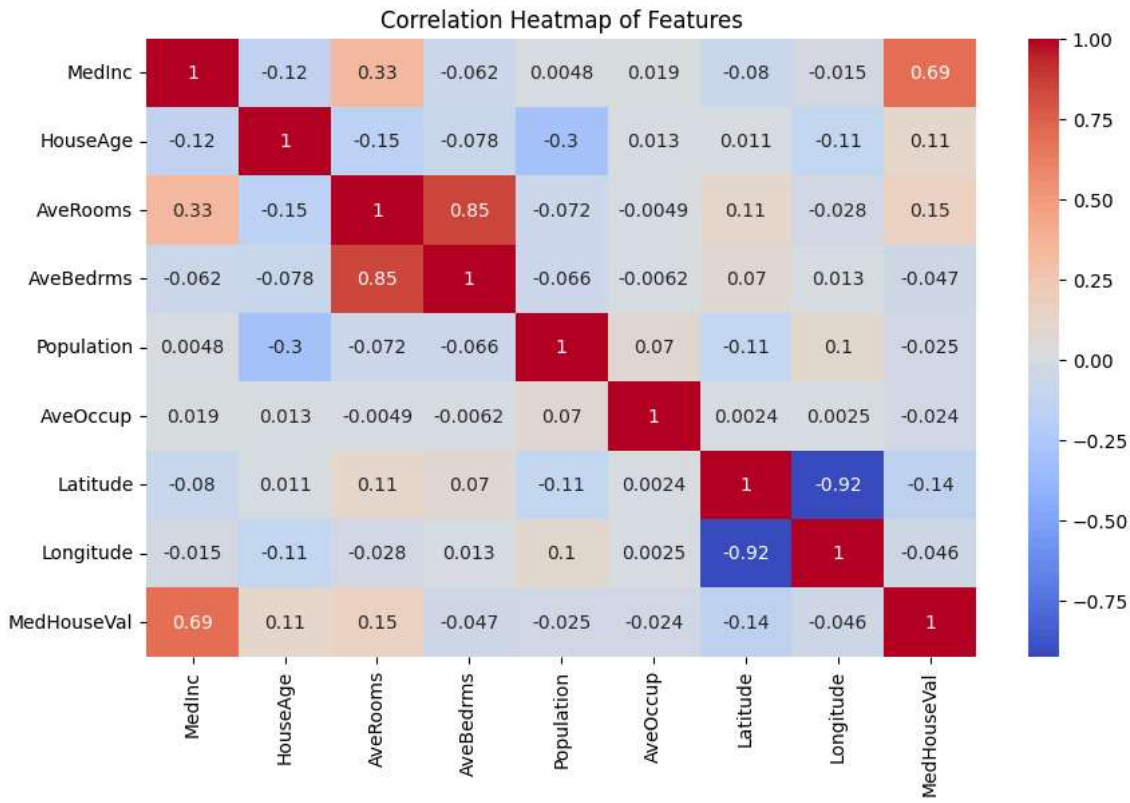
```
data.describe()
```

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal
count	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000
mean	3.870671	28.639486	5.429000	1.096675	1425.476744	3.070655	35.631861	-119.569704	2.068558
std	1.899822	12.585558	2.474173	0.473911	1132.462122	10.386050	2.135952	2.003532	1.153956
min	0.499900	1.000000	0.846154	0.333333	3.000000	0.692308	32.540000	-124.350000	0.149990
25%	2.563400	18.000000	4.440716	1.006079	787.000000	2.429741	33.930000	-121.800000	1.196000
50%	3.534800	29.000000	5.229129	1.048780	1166.000000	2.818116	34.260000	-118.490000	1.797000
75%	4.743250	37.000000	6.052381	1.099526	1725.000000	3.282261	37.710000	-118.010000	2.647250
max	15.000100	52.000000	141.909091	34.066667	35682.000000	1243.333333	41.950000	-114.310000	5.000010

```
print("\nMissing Values:\n", data.isnull().sum())
```

```
Missing Values:
 MedInc         0
HouseAge       0
AveRooms       0
AveBedrms      0
Population     0
AveOccup       0
Latitude       0
Longitude      0
MedHouseVal    0
dtype: int64
```

```
plt.figure(figsize=(10,6))
sns.heatmap(data.corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap of Features')
plt.show()
```




```
X = data.drop('MedHouseVal', axis=1)
y = data['MedHouseVal']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
print(X_train_scaled.shape)
print(X_test_scaled.shape)
```

```
(16512, 8)
(4128, 8)
```

```
print(X_train_scaled)
```

```
[[-0.326196    0.34849025 -0.17491646 ...  0.05137609 -1.3728112
   1.27258656]
 [-0.03584338  1.61811813 -0.40283542 ... -0.11736222 -0.87669601
   0.70916212]
 [ 0.14470145 -1.95271028  0.08821601 ... -0.03227969 -0.46014647
  -0.44760309]
 ...
```

```
 [-0.49697313  0.58654547 -0.60675918 ...  0.02030568 -0.75500738
   0.59946887]
 [ 0.96545045 -1.07984112  0.40217517 ...  0.00707608  0.90651045
  -1.18553953]
 [-0.68544764  1.85617335 -0.85144571 ... -0.08535429  0.99543676
  -1.41489815]]
```

```
model = LinearRegression()
model.fit(X_train_scaled, y_train)
```

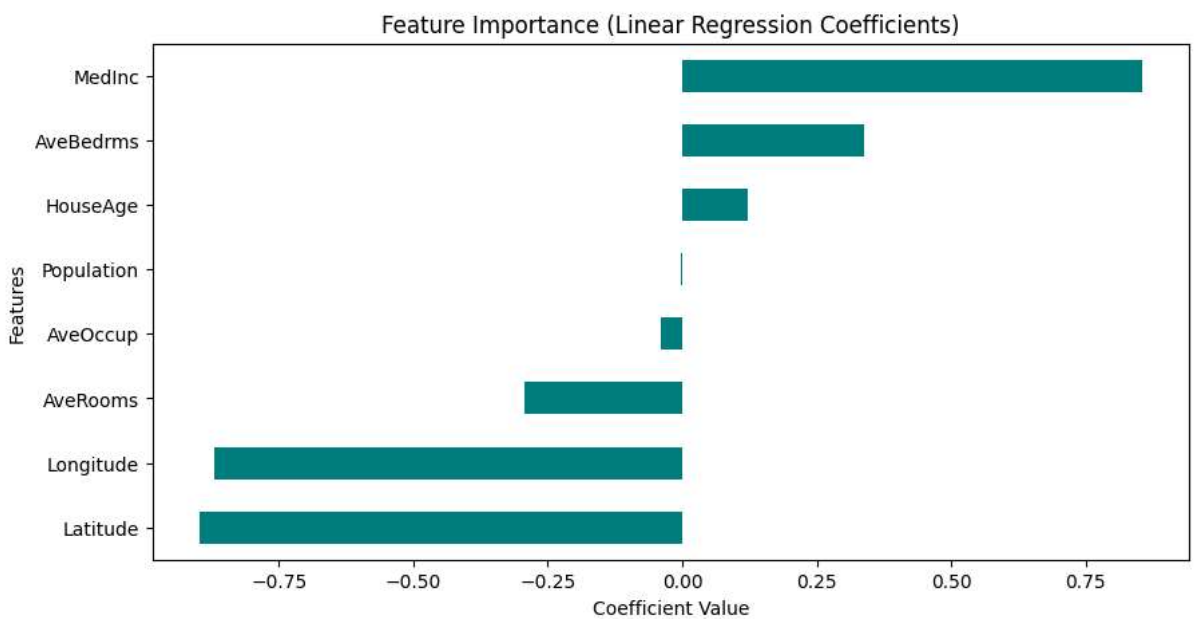
```
▾ LinearRegression
LinearRegression()
```

```
y_pred = model.predict(X_test_scaled)
```

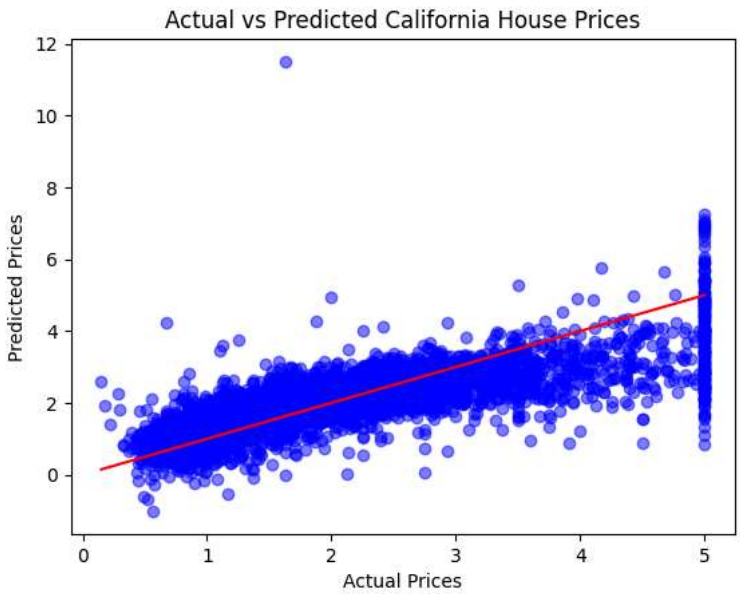
```
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("\nIntercept:", model.intercept_)
print("Coefficients:", model.coef_)
print("Mean Squared Error:", mse)
print("R² Score:", r2)
```

```
Intercept: 2.0719469373788777
Coefficients: [ 0.85438303  0.12254624 -0.29441013  0.33925949 -0.00230772 -0.0408291
 -0.89692888 -0.86984178]
Mean Squared Error: 0.5558915986952442
R² Score: 0.575787706032451
```

```
feature_importance = pd.Series(model.coef_, index=X.columns)
plt.figure(figsize=(10,5))
feature_importance.sort_values().plot(kind='barh', color='teal')
plt.title('Feature Importance (Linear Regression Coefficients)')
plt.xlabel('Coefficient Value')
plt.ylabel('Features')
plt.show()
```




```
plt.scatter(y_test, y_pred, color='blue', alpha=0.5)
plt.plot([min(y_test), max(y_test)], [min(y_test), max(y_test)], color='red')
plt.xlabel('Actual Prices')
plt.ylabel('Predicted Prices')
plt.title('Actual vs Predicted California House Prices')
plt.show()
```




```
sample_input = np.array([[8, 20, 5, 1, 1000, 3, 34, -118]])
sample_input_scaled = scaler.transform(sample_input)
predicted_value = model.predict(sample_input_scaled)[0]
print(f"\nPredicted Median House Value: ${predicted_value * 100000:.2f}")
```

```
Predicted Median House Value: $381913.35
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but
  warnings.warn(
```